

Session 13: Create a script using AWS CLI to convert a text file into speech using Polly

Exp-12: AWS Polly – Step-by-Step Procedure (Starting from Polly Usage)

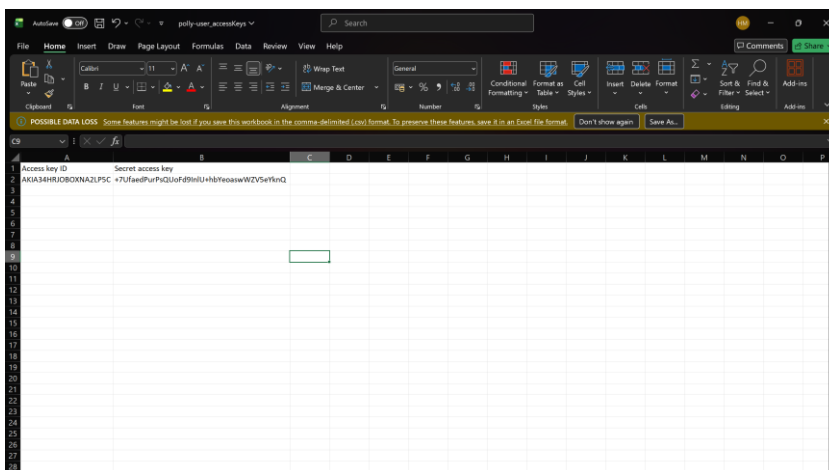
Step 1: Open AWS Polly (Verification Step)

1. Go to AWS Console
2. Search → **Polly**
3. Open **Amazon Polly**
4. Enter sample text:
Hello AWS Polly Test
5. Select voice:
 - Joanna (English)
 - Aditi (Indian)
6. Click **Listen to speech**

✓ Confirms Polly service is working

Generate Access Keys:

1. Open the created user → go to **Security credentials**
2. Scroll to **Access keys**
3. Click **Create access key**
4. Choose:
 - ✓ Command Line Interface (CLI)
5. Click **Next → Create**
6. **Download or copy:**
 - Access Key ID
 - Secret Access Key
7. Download The Csv File



Install and Configure AWS CLI

Step 1: Search -> Cloud Shell

- pip install awscli

Check installation:

- aws --version

Step 2: Configure AWS

Run:

- `aws configure`

Enter details:

- AWS Access Key ID: XXXXX
- AWS Secret Access Key: XXXXX
- Default region name: us-east-1
- Default output format: json

```
~ $ aws configure
AWS Access Key ID [None]: AKIA34HRJOBXNA2LP5C
AWS Secret Access Key [None]: +7UfaedPurPsQUoFd9In1U+hbYeoaswWZV5eYknQ
Default region name [None]: eu-north-1
Default output format [None]: json
~ $
```

✓ CLI is now connected to AWS

Step 3: Install Required Library

- `pip install boto3`

Step 4: Create Python Program

Create file: `nano polly_tts.py`

Paste Code:

```
import boto3
# Create Polly client
polly = boto3.client('polly', region_name='us-east-1')

# Input text
text = "Hello! Welcome to AWS Polly text to speech demo."

# Convert text to speech
response = polly.synthesize_speech(
    Text=text,
    OutputFormat='mp3',
    VoiceId='Joanna'
)

# Save audio file
if "AudioStream" in response:
    with open("speech.mp3", "wb") as file:
        file.write(response['AudioStream'].read())
    print("Speech file saved successfully!")
else:
    print("Error generating speech")
```

```
GNU nano 8.3 polly_tts.py
import boto3

# Create Polly client
polly_client = boto3.client('polly', region_name='eu-north-1')

# Input text
input_text = "Hello! Welcome to AWS Polly text to speech demo."

# Request speech synthesis
response_data = polly_client.synthesize_speech(
    Text=input_text,
    OutputFormat='mp3',
    VoiceId='Joanna'
)

# Save output file
if 'AudioStream' in response_data:
    with open("speech.mp3", "wb") as audio_file:
        audio_file.write(response_data['AudioStream'].read())
    print("Speech file saved successfully!")
else:
    print("Error generating speech")
```

Save And Exit The Program

->Then Click ctrl + o Enter
Click ctrl + x

Step 5: Run the Program

python polly_tts.py

```
~ $ nano polly_tts.py
~ $ python polly_tts.py
/usr/local/lib/python3.9/site-packages/boto3/compat.py:89: PythonDeprecationWarning: Boto3 will no longer support Python 3.9 starting April 29, 2026. To continue receiving service updates, bug fixes, and security updates please upgrade to Python 3.10 or later. More information can be found here: https://aws.amazon.com/blogs/developer/python-support-policy-updates-for-aws-sdks-and-tools/
  warnings.warn(warning, PythonDeprecationWarning)
Speech file saved successfully!
~ $
```

Step 6: Verify Output

ls

Output:

speech.mp3

```
~ $ ls
polly_tts.py  speech.mp3
~ $
```

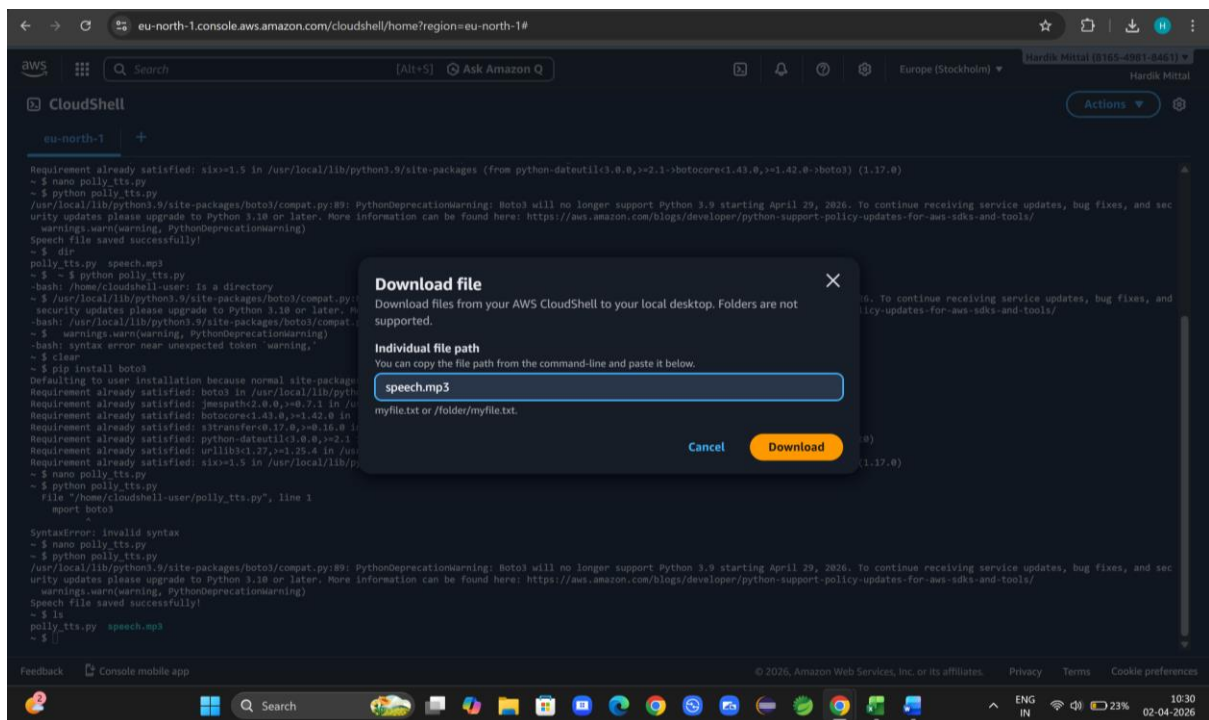
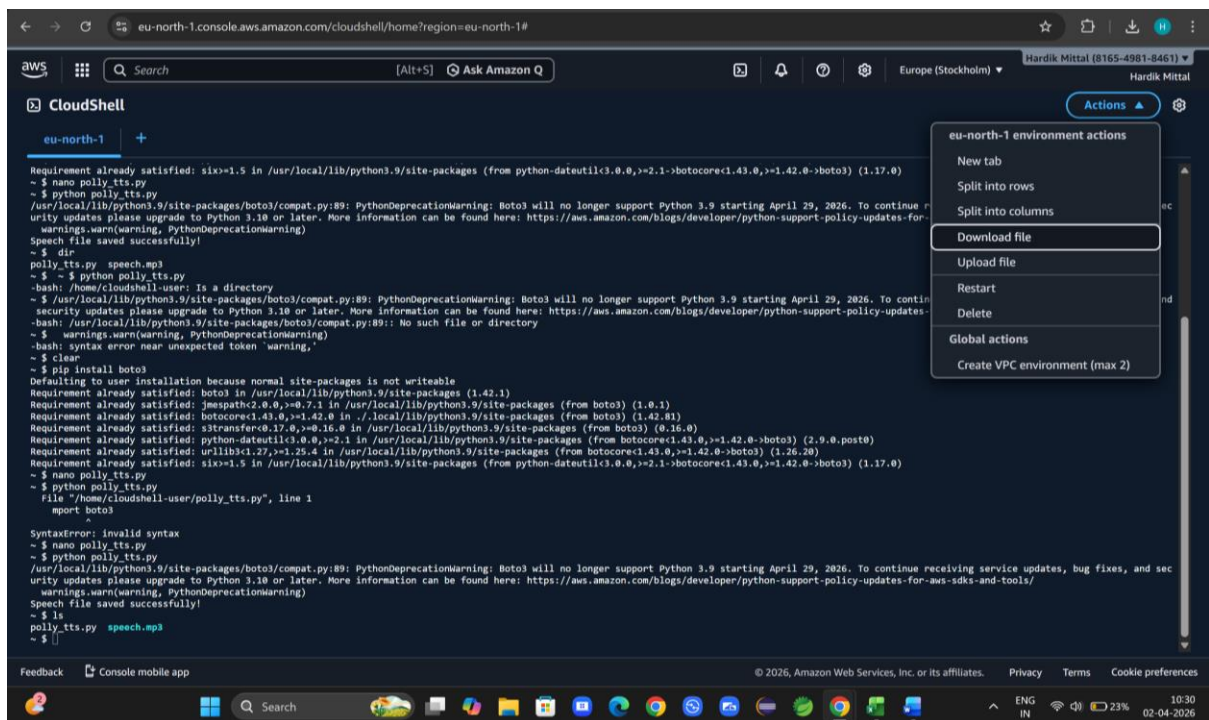
◆ Step 7: Download File (Important for CLI Users)

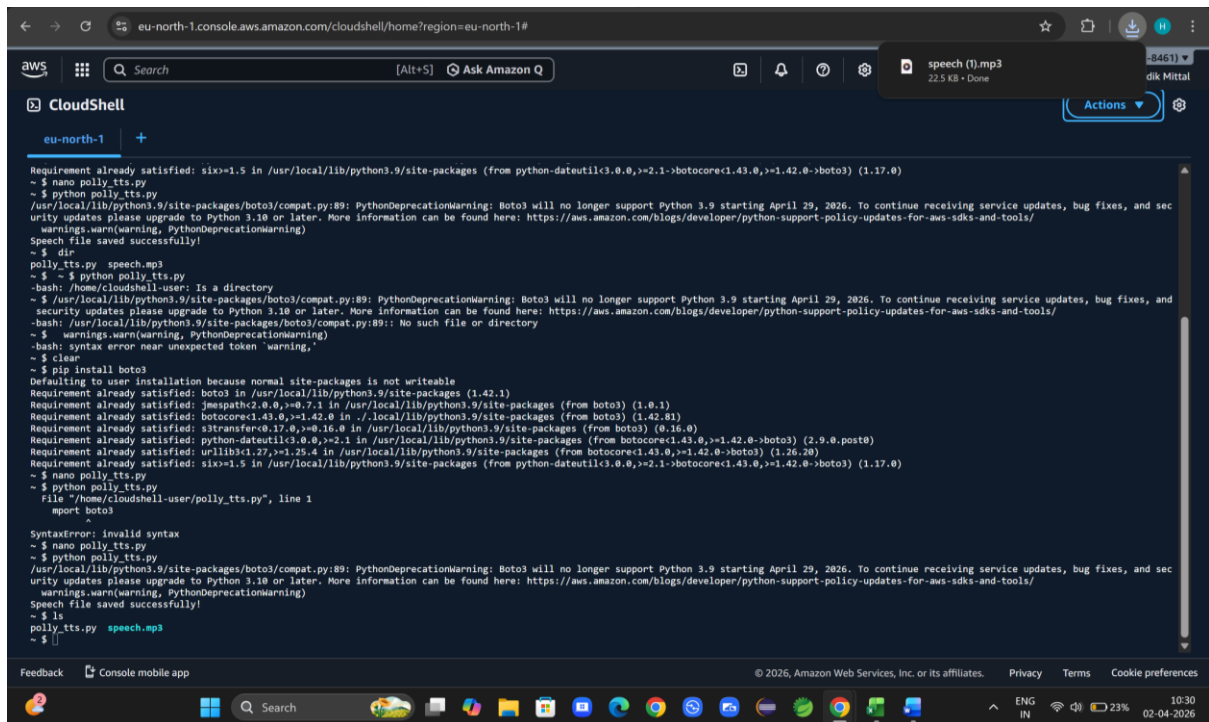
📁 If using AWS CloudShell:

1. Click **Actions** → **Download file**
2. Enter:

speech.mp3

📁 File downloads to your system





```
eu-north-1 console.aws.amazon.com/cloudshell/home?region=eu-north-1#
AWS CloudShell
eu-north-1
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.9/site-packages (from python-dateutil<3.0.0,>=2.1->botocore<1.43.0,>=1.42.0->boto3) (1.17.0)
~ $ nano polly_tts.py
~ $ python polly_tts.py
/usr/local/lib/python3.9/site-packages/boto3/compat.py:89: PythonDeprecationWarning: Boto3 will no longer support Python 3.9 starting April 29, 2026. To continue receiving service updates, bug fixes, and security updates please upgrade to Python 3.10 or later. More information can be found here: https://aws.amazon.com/blogs/developer/python-support-policy-updates-for-aws-sdks-and-tools/
warnings.warn(warning, PythonDeprecationWarning)
Speech file saved successfully!
~ $ dir
polly_tts.py  speech.mp3
~ $ ~ $ python polly_tts.py
-bash: /home/cloudshell-user: Is a directory
~ $ /usr/local/lib/python3.9/site-packages/boto3/compat.py:89: PythonDeprecationWarning: Boto3 will no longer support Python 3.9 starting April 29, 2026. To continue receiving service updates, bug fixes, and security updates please upgrade to Python 3.10 or later. More information can be found here: https://aws.amazon.com/blogs/developer/python-support-policy-updates-for-aws-sdks-and-tools/
warnings.warn(warning, PythonDeprecationWarning)
-bash: /usr/local/lib/python3.9/site-packages/boto3/compat.py:89: No such file or directory
~ $ ~ $ warnings.warn(warning, PythonDeprecationWarning)
-bash: syntax error near unexpected token 'warning,'
~ $ clear
~ $ pip install boto3
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: boto3 in /usr/local/lib/python3.9/site-packages (1.42.1)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /usr/local/lib/python3.9/site-packages (from boto3) (1.0.1)
Requirement already satisfied: botocore<1.43.0,>=1.42.0 in ./local/lib/python3.9/site-packages (from boto3) (1.42.81)
Requirement already satisfied: s3transfer<0.17.0,>=0.16.0 in /usr/local/lib/python3.9/site-packages (from boto3) (0.16.0)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /usr/local/lib/python3.9/site-packages (from botocore<1.43.0,>=1.42.0->boto3) (2.9.0.post0)
Requirement already satisfied: urllib3<1.27,>=1.25.4 in /usr/local/lib/python3.9/site-packages (from botocore<1.43.0,>=1.42.0->boto3) (1.26.20)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.9/site-packages (from python-dateutil<3.0.0,>=2.1->botocore<1.43.0,>=1.42.0->boto3) (1.17.0)
~ $ nano polly_tts.py
File "/home/cloudshell-user/polly_tts.py", line 1
import boto3
^
SyntaxError: invalid syntax
~ $ nano polly_tts.py
~ $ python polly_tts.py
/usr/local/lib/python3.9/site-packages/boto3/compat.py:89: PythonDeprecationWarning: Boto3 will no longer support Python 3.9 starting April 29, 2026. To continue receiving service updates, bug fixes, and security updates please upgrade to Python 3.10 or later. More information can be found here: https://aws.amazon.com/blogs/developer/python-support-policy-updates-for-aws-sdks-and-tools/
warnings.warn(warning, PythonDeprecationWarning)
Speech file saved successfully!
~ $ ls
polly_tts.py  speech.mp3
~ $
```

Step 8: Play Audio

- Open downloaded file
- Play using:
 - VLC Player
 - Windows Media Player